Int. J. of Precis. Eng. and Manuf.-Green Tech.
https://doi.org/10.1007/s40684-025-00833-x

Online ISSN 2198-0810
Print ISSN 2288-6206

REGULAR PAPER

A Novel Hybrid Lubri-Cooling Approach for Machining of Ti-6Al-4V Through Combination of Pulsated Minimal Quantity Lubrication and Dry Ice Media

Muhammad Ahmed Khan[1,2,3] · Bing Wang[1,2,3] · Zhanqiang Liu[1,2,4] · Aqib Mashood Khan[5] · B. Mouli Prasanth[1,2,3] · Aisheng Jiang[6]

Received: 28 May 2025 / Revised: 28 November 2025 / Accepted: 10 December 2025
© The Author(s), under exclusive licence to Korean Society for Precision Engineering 2025

Abstract

Enhancing the machinability of Ti-6Al-4V titanium alloy under high-speed face milling remains a significant challenge due to its low thermal conductivity and high chemical reactivity. In this study, a novel eco-efficient hybrid cooling/lubrication strategy combining dry ice (i.e., solid state CO_2) assisted cooling with a pulsated minimum quantity lubrication (MQL) system was investigated. Different with conventional MQL or cryogenic approaches, the proposed system utilizes a low MQL flow rate of 35 mL/h integrated with dry ice cooling to ensure simultaneous lubrication and enhanced thermal control at the cutting interface. Milling experiments were performed under three different conditions as dry, pulsated MQL, and hybrid (pulsated MQL + dry ice) at varying cutting speeds of 100–250 m/min) and feed rates of 0.08–0.12 mm/z. The hybrid system's performance was evaluated through comprehensive analysis of tool wear, cutting force, machined surface roughness and topography, phase composition, and subsurface microhardness. The results demonstrated that the hybrid environment significantly reduced tool wear and cutting forces while improving surface quality and preserving the material's crystalline structure. Machined surface roughness R_a was reduced by up to 83% compared to dry machining, and X-ray diffraction analysis revealed minimal formation of secondary phases under hybrid conditions. Furthermore, the hybrid setup showed lower subsurface microhardness, indicating effective suppression of thermally induced strain hardening. Overall, the proposed hybrid MQL-dry ice cooling strategy proved highly effective in improving machining performance and surface integrity while aligning with sustainable manufacturing goals through reduced lubricant usage.

Keywords Hybrid cooling/Lubrication · Ti-6Al-4V machining · Pulsated minimum quantity lubrication · Dry ice · Surface integrity

✉ Bing Wang
sduwangbing@sdu.edu.cn

1 School of Mechanical Engineering, Shandong University, Jinan 250061, China

2 State Key Laboratory of Advanced Equipment and Technology for Metal Forming, Shandong University, Jinan 250061, China

3 Key Laboratory of High Efficiency and Clean Mechanical Manufacture of MOE/Shandong key Laboratory of High Performance Tools and System, Jinan 250061, China

4 School of Airspace Science and Engineering, Shandong University, Weihai 264209, China

5 College of Mechanical Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing 210016, China

6 Zhuzhou Cemented Carbide Cutting Tools Co., Ltd, Zhuzhou 412007, China

1 Introduction

Ti-6Al-4V is known as a material of utmost importance in aerospace components, cardiovascular surgical devices, and chemical industrial piping systems due to its excellent physical and mechanical properties. This titanium alloy possesses excellent strength-weight ratio, outstanding corrosion resistance, and high thermal stability. Therefore, Ti-6Al-4 V is highly suitable for high-performance applications in aircraft structures and for advanced biomedicine or engineering [1]. It effectively contributes to the performance and efficiency improvement of modern aircraft, which is critically important for fuel economy and the durability of commercial and military aviation applications [2]. However, the excellent properties of Ti-6Al-4 V make it face great challenges

Published online: 22 December 2025

during machining process with dry cutting condition. The poor heat conductivity, high strength and strong capability of machinability for cutting tools of such titanium alloy make machining difficult, and there are prominent issues of tool wear, cutting efficiency and surface integrity during machining [3, 4]. In order to alleviate these problems, the flood-cutting method using large amount of cutting fluid has been used. Cutting liquids serve to reduce the tool wear, evacuate heat from the cutting area and lubricate the interface, which is then beneficial to improve surface finish and provide lubrication along with enhanced heat dissipation during machining [5]. Nevertheless, the extensive use of cutting fluids involves important environmental and health concerns. Most cutting fluids are oil-based, and their dumping can pose a threat to water and soil pollution [6]. In addition, the production, delivery, and disposal of these fluids contribute significantly to the carbon footprint of machining operations and are considered to be a challenge to sustainability [7]. Furthermore, vaporization of cutting fluids in the machining process creates harmful aerosols, which pose respiratory and dermal exposure to workers and thus raise critical occupational health issues [8].

To overcome the problems associated with dry flood cutting, Minimum Quantity Lubrication (MQL) using vegetable oils as cutting fluids has emerged as an environmentally healthier solution [9]. MQL is a minimal quantity lubrication machining process, which significantly reduces the amounts of cutting fluid used in operations [10–12]. The decreased fluid consumption therefore makes it an environment-friendly approach, which not only has the potential to save waste disposal costs, but is environmentally friendly and also enhances machining efficiency [13]. Iyappan and Ghosh [14] found that the application of MQL in aluminum alloy machining could reduce the cutting friction coefficient of uncoated end mill and prevent the formation of built-up edge. Similarly, Vazquez et al. [15] analyzed the influence of cooling and lubricating conditions on machining of Ti-6Al-4 V uncovering that MQL increased the quality of the surface, enhanced the precision of machining, and reduced the wear tool. Finally, MQL is in accordance with green machining philosophy working towards environmentally sustainable machining. In a study conducted by Hadad and Sadeghi [16] on turning of 4140 steel, enhanced performance of MQL method over dry cutting and conventional coolant were expressed. Better cutting force, temperature management, and surface quality in MQL make it as good alternative to enhance the machining efficiency. Sarikaya et al. [17] and He et al. [18] described the benefits and challenges of MQL technology in various industries. They determined that MQL mitigates the shortcomings of traditional lubrication systems including inadequate cooling and lubrication, and significantly contributes to the advancement of sustainable manufacturing practices.

New atomization cutting fluid delivery systems have been developed recently that utilize atomized conventional cutting fluids for oil delivery directly to the tool-workpiece interface [19]. Nath et al. [20, 21] carried out a series of studies for comparison between atomization-based cutting spray system and traditional coolants or dry air so as to enhance the machinability on titanium alloys. They found that atomized cutting sprays significantly increased tool life, reduced machining costs, and improved surface finish. In a similar vein, Roy et al. [22] studied a new pulse-mode MQL method in turning of 4340 steel to analyze the principal and secondary flank wear, surface roughness, surface topology, chip shape, and chip reduction coefficient. Their findings contribute to the advancement of cutting processes aiming to improve performance and sustainability.

In cryogenic machining, emulsion coolants are substituted with gases, liquids, or solids that markedly lower the temperature in the vicinity of the cutting zone. Conventional "sub-zero" cooling techniques, including liquid nitrogen (LN_2) and solid or liquid carbon dioxide (CO_2), can improve the machining process, especially in scenarios where elevated cutting temperatures are anticipated [23, 24]. Zhao et al. [25] performed milling experiments on Ti-6Al-4 V alloy utilizing liquid nitrogen cooling, yielding enhanced surface quality relative to direct milling without cooling. It was determined that liquid nitrogen cooling augmented residual compressive stress on the machined surface that can enhance machined surface quality. Similarly, Jebaraj et al. [23] investigated end milling of Al 6082-T6 alloy in cryogenic CO_2, LN_2 and wet states. They reported that cryogenic LN_2 had the lowest cutting temperatures which were 38.29% and 32.8% lower than wet and CO_2 cooling, respectively. However, traditional coolant supplied obtained better machined surface roughness with certain cutting parameters. Sivaiah and Chakradhar [26] conducted turning studies on 17 − 4 PH stainless steel and discovered that cryogenic machining surpassed dry, wet, and minimum quantity lubrication conditions regarding tool wear, surface finish, and chip morphology. This underscores the potential of cryogenic cooling in enhancing machining performance and surface quality.

The integration of supercritical CO_2 (SC-CO_2) cooling with MQL technology has attracted significant interest from researchers investigating its effects on machining performance [27]. Studies have shown that this integrated system, when utilized for machining AISI 304 L, can attain machined surface roughness and residual stress levels comparable to those achieved with flood coolant milling. Meanwhile, it provides a markedly elevated material removal rate [28]. Additionally, for machining of titanium

Table 1 Chemical composition (wt%) of Ti-6Al-4 V alloy as specified by the manufacturer

Titles	Al	V	Fe	C	*N*	H	O	Others	Ti
Ti-6Al-4 V	6.32	4.0	0.14	0.01	0.01	≤0.015	≤0.20	≤0.4	Balance

Table 2 Physical and mechanical properties of Ti-6Al-4 V alloy as provided by the manufacturer

Density (kg/m^3)	Hardness (HRC)	Tensile strength (MPa)	Yield strength (MPa)	Elongation (%)
4428	36	1275	1187	11.5

alloys particularly at elevated cutting speeds, the integration of MQL and SC-CO_2 cooling has been shown to significantly diminish cutting forces [27]. Jamil et al. [29] validated the efficacy of the cryogenic CO_2-snow technique in milling titanium alloys, and determined that the machinability performance is ranked as CO_2-snow > flood cooling > subzero MQL. These findings underscore the prospective advantages of employing SC-CO_2 and MQL to enhance efficiency and diminish forces in high-performance machining applications.

As mentioned, to tackle the challenges of increasing cutting velocity, feed rate and material removal rate in the process of machining difficult-to-cut material Ti-6Al-4 V is of industrial importance. In contrast to the conventional continuous MQL or cryogenic methods, the novel hybrid cooling and lubrication approach features an integrated dry ice assisted cooling and a pulsated MQL system with a low oil flow rate of 35mL/h, which enables efficient reduction in lubricant consumption while ensuring the required lubrication at the interface. The purpose of this study is to explore the eco-efficient hybrid system in high-speed face milling of Ti-6Al-4 V to study its performance. For comparison purpose, identical milling experiments were carried out under dry and pulsated MQL conditions. The research systematically evaluates important machining performance measures such as cutting forces, tool wear, machined surface and subsurface integrity to understand the overall efficacy of the developed cooling approach.

2 Experimental Procedures

2.1 Materials, Cutting Tools, and Machine Tool

The experimental study employed rectangular workpieces composed of Ti-6Al-4 V alloy with dimensions of 100 mm × 80 mm × 25 mm. The chemical compositions and physical-mechanical properties of the Ti-6Al-4 V alloy, as provided by the manufacturer, are summarized in Tables 1 and 2, respectively.

In the milling experiments, uncoated carbide milling inserts (Zhuzhou YW2, model 4160511, size of 16 mm × 16 mm× 4.5 mm) with four cutting edges were employed with a rake angle of 11°, clearance angle of 11°, major cutting-edge angle of 75°, and nose radius of 0.50 mm. Such cutting inserts were selected to provide stable edge geometry and representative performance for face milling of TC4 under dry, pulsated MQL, and hybrid (MQL + dry-ice CO_2) conditions. The cutting tools were fixed to an eight-slotted tool holder with a diameter of 125 mm and connected to a high-speed vertical milling machine. Two cutting inserts were employed simultaneously, with each experimental condition involving the use of a new cutting edge for both tools and all tests were conducted with three repetitions per condition. The selection of four cutting speeds (100–250 m/min) and two feed rates (0.08 and 0.12 mm/rev) was based on typical industrial practices for milling Ti–6Al–4 V with uncoated carbide tools [30]. These cutting parameters are representative of both conventional and high-speed machining conditions, allowing for a detailed assessment of the impact of different cooling and lubrication methods on tool wear, cutting forces, and overall machining performance. The chosen parameters ensure a broad evaluation of the thermal and mechanical factors that influence wear mechanisms in the machining process. The Ti-6Al-4 V workpiece was firmly affixed to the center of the high-speed vertical milling machine utilizing a precise vise. The studies were performed on a Shenyang YMC0540d high-speed vertical milling machine, which can attain a maximum spindle speed of 30,000 rpm and has a worktable dimension of 400 mm X 600 mm. Additional information concerning the experimental configuration and the parameters of the milling experiments is illustrated in Fig. 1.

2.2 Pulsating MQL System

A specialized MQL system is used to produce the pulse MQL environment during the machining of Ti-6Al-4 V in this work. MQL system sprays fogging droplets of oil and air on the precise cutting area. In the current experiments, the continuous supply of the lubricant in conventional MQL mode system was turned into pulse to provide control over the timing and frequency of the air-lubricant mist jet expulsion on the cut zone. The aim of applying pulse MQL is to reduce the consumption of lubrication to a lower level and observe its effect on machining performance.

The MQL system functions with a 24 V DC power supply, employing compressed air at a pressure of 4 bar. The vegetable oil utilized in the system possesses a viscosity of 22 cSt at 40 °C, a flash point of 260 °C, and a pour point of

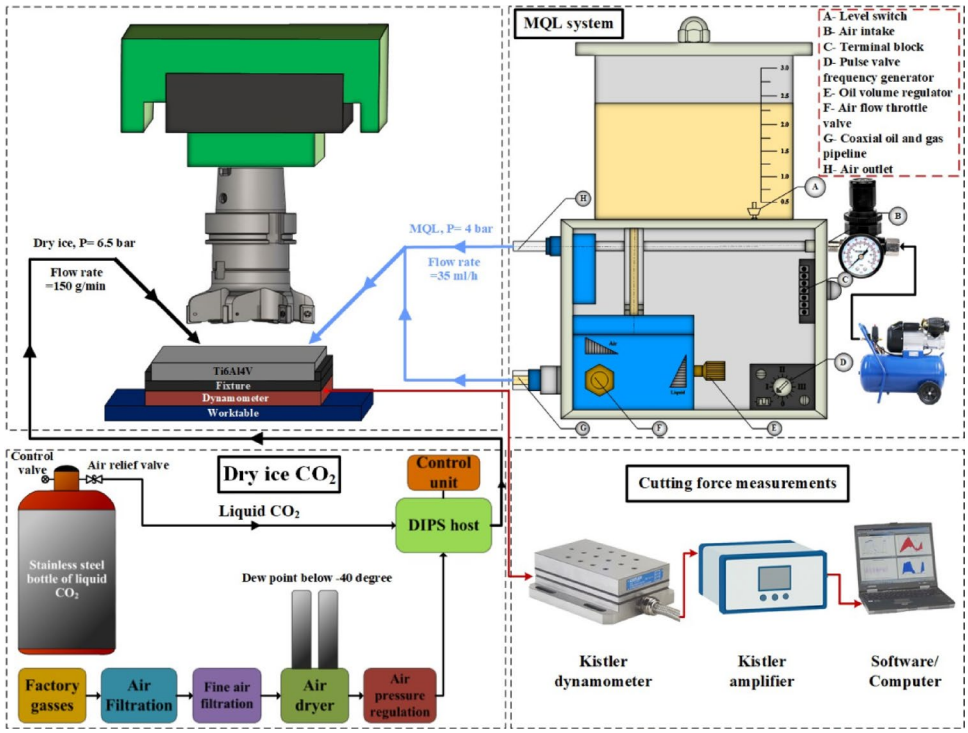

Fig. 1 Experimental setup used for machining of Ti-6Al-4 V with hybrid lubri-cooling approach

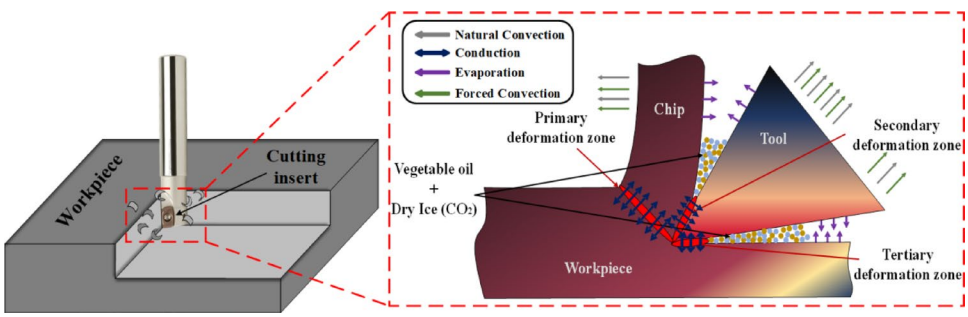

Fig. 2 Schematic representation of heat transfer and lubrication mechanism under hybrid lubrication machining conditions

−30 °C, rendering it suitable for high-performance machining by ensuring efficient lubrication and cooling. Lubrication is facilitated by means of a pneumatic dredge pump controlled by a frequency generator. The system can be adjusted to vary the frequency of the pulses and the amount of oil dispensed per pulse. The pump can be set to deliver up to 0.035 ml of oil per pulse, and the pulse rate can be controlled to alter the volume of oil being discharged. More precisely, the system works with a 2 s pulse cycle that sees the pump on for 2 s, during which it is supplying oil, and off for the subsequent 2 s. This pulsating system ensures precise control of the oil fed up to the cutting area, with the system provider, 35 ml/h of oil at that pulse rate.

One of the main features of this MQL system is the presence of coaxial pipes, designed to deliver oil and compressed air separately to the cutting zone. Its coaxial setup ensures the oil and air each move down different passageways to avoid premature blending before reaching the cutting point. The external pipe (Ø = 8 mm) carries the compressed air, and the internal pipe (Ø = 3 mm) carries vegetable oil. The double-pipe system facilitates the efficient lubricating, the vegetable oil plays a role of lubricant between cutting tool and workpiece and the compressed air helps to remove heat generated and chips in the cutting area. In addition to the experimental setup, Fig. 2. shows the method of cooling and lubrication used in the machining process.

2.3 Dry Ice Setup and Experimental Parameters

In the present study, an implementation of a dry ice setup (model: DIPS) was used to introduce dry ice particles to the cutting zone of titanium machining for the cooling purpose. The device functions by initially extracting siphon-type liquid CO_2 from 40 kg cylinders to guarantee a steady CO_2 supply. Upon exiting the nozzle, the liquid CO_2 experiences rapid expansion due to Joule-Thomson effect, converting into dry ice particles that are directly supplied to the cutting region for efficient cooling.

The system functions at a pressure of 6.5 bar prior to injection, guaranteeing that the CO_2 is supplied under elevated pressure for best efficacy. The flow rate is established at 150 g/min, ensuring a uniform and regulated discharge of dry ice particles into the machining area. The dry ice technology brings cooling, embrittlement and sublimation phenomena to the adhesive particles, so that these types of particles to be in an unstable state. Hence, the small dry ice particles are convoyed to efficiently prevent chip adhesion, welding, and large friction between chip and cutting tool.

A main component of the system is its control unit, which manages the CO_2 flow and oversees the pressure to guarantee optimal functionality. Auxiliary gases, commonly air, are used with CO_2 to facilitate cooling and chip removal, hence enhancing process efficiency. The nozzle is engineered to dispense CO_2 and auxiliary gases with precision, guaranteeing that dry ice particles are accurately guided to the required locations. The milling test parameters for Ti-6Al-4V material are presented in Table 3.

2.4 Measurements and Methodological Approach

A Japan Electronics JMS-7610 F field emission scanning electron microscope (FE-SEM) was employed to investigate the tool wear mechanisms on both the flank and rake faces and workpiece machined surfaces under the highest applied machining parameters. Surface topography of the machined specimens was examined using a KEYENCE VK-S1000K/S1100 profile analyzing laser microscope. Axial cutting forces were measured using a Kistler 9190AA dynamometer connected to a Kistler control unit, with data acquisition and analysis performed via Dynoware software. Surface roughness R_a was evaluated using a TR200 Handheld Roughness Tester, with measurements taken at three distinct positions start, middle, and end along the machined surface. The average of these three values was used for analysis. Surface topography was further assessed using optical microscopy to observe surface texture and feed mark patterns.

Microhardness evaluations were conducted using an HV-1000 A Micro Vickers Hardness Tester under a 0.5 kgf load and a dwell time of 15 s. To prepare for subsurface measurements, the cross-sections of the machined samples were sequentially ground using abrasive to fine-grade sandpapers and then polished with polishing agent at 150 rpm on MP-3 S Metallographic Automatic Grinding and Polishing Machine until a mirror finish was achieved. Following preparation, microhardness was measured from the surface down to a depth of 50 µm, at intervals of 10 µm, to capture the hardness gradient beneath the machined surface.

Table 3 Machining parameters for Ti-6Al-4 V under different environmental conditions

Parameters	Description
Machine tool	• CNC horizontal milling machine
Cutting tool	• Uncoated carbide milling inserts (Zhuzhou YW2-4160511); number of edges = 4; cutting edge radius = 0.5 mm; clearance/rake angle = 11°
Workpiece	• Ti-6Al-4 V grade 5 (100 mm × 80 mm × 25 mm)
Milling parameters	• Cutting speed V_c (m/min): 100, 150, 200, 250 • Feed rate f_z (mm/rev): 0.08, 0.12 • Axial depth of cut a_p (mm): 0.8 • Radial depth of cut a_e (mm): 8
Environments	• Dry, Pulsated MQL and Pulsated MQL + Dry Ice
Vegetable based oil	• Viscosity at 40 °C: 22 cSt • Pour point: −30 °C • Flash point: 260 °C
Pulsated MQL settings	• Vegetable oil • Flowrate: 35 ml/h • Pressure: 4 bar • Nozzle diameter: 2.5 mm • Nozzle to cut distance and angle: 30 mm, 30°
Dry Ice	• Siphon type CO_2 • Flowrate: 150 g/min • Pressure: 6.5 bar • Nozzle diameter: 2 mm • Nozzle to cut distance and angle: 30 mm, 30°

3 Results and Discussion

3.1 Tool Wear Analysis

During milling operations, cutting conditions have a significant impact on tool degradation, particularly when machining hard-to-cut materials like Ti-6Al-4 V, where the tool is subjected to elevated thermal and mechanical loads. Parameters such as the cutting environment, machining settings, workpiece material, tool material, and tool geometry especially at the cutting edge directly influence tool wear behavior [31]. The dominant wear mechanisms observed include flank wear and crater wear, which are analyzed in the following sections. The tool wear assessments were conducted under milling conditions comprising a cutting speed of 250 m/min, axial depth of cut of 0.8 mm, radial depth of cut of 8 mm, and feed rate of 0.12 mm/rev. Each experiment involved a cutting length of 100 mm, and the same tool was used for three repetitions under the same conditions. After completing the three repetitions, the mechanism of tool wear was analyzed.

3.1.1 Tool Flank Wear

Flank wear is a common mode of tool degradation during milling operations; however, its severity varies significantly with the cutting environment. Among the three tested conditions, dry machining exhibited the most severe flank wear, as shown in Fig. 3. SEM analysis of the worn tool revealed dominant adhesion, debris entrapment, and chipping along the flank face. In addition, localized notching and the presence of microcracks were also observed, indicating complex wear interactions under dry conditions. The absence of external cooling and lubrication in dry machining leads to elevated temperatures at the tool-workpiece interface, which promotes adhesive wear. This occurs as a result of the strong chemical affinity between the titanium alloy and the tool material, especially under high thermal and mechanical loads [32]. The adhered material can become unstable and contribute to debris entrapment, where fragmented material accumulates between the cutting edge and the workpiece. The occurrence of chipping along the cutting edge can be attributed to repeated mechanical impacts and high cyclic stresses during tool engagement with the workpiece, which progressively diminish the fracture resistance of the tool material. These localized impacts contribute to the initiation and propagation of microcracks, likely driven by thermal fatigue. In the absence of adequate cooling, the accumulated thermal and mechanical loads promote crack growth, ultimately accelerating tool degradation [33].

In addition to chipping and microcracking, localized notching was also observed, particularly near the depth-of-cut region. Although less dominant, this form of wear is associated with the hardened surface layer of the workpiece, elevated thermal gradients, and potential oxidation effects. The inability to dissipate heat efficiently under dry conditions exacerbates structural damage along the tool edge, further contributing to the overall wear severity.

The EDS elemental mapping conducted on worn tool surfaces both on the flank and crater regions confirmed the presence of titanium and its alloying elements, indicating material transfer from the workpiece to the tool. Due to the absence of lubrication or cooling, friction between the tool and workpiece increases, leading to higher temperatures at the cutting interface. This promotes tool wear primarily through adhesive and abrasive mechanisms, where material from the workpiece is more likely to be worn away than transferred to the tool. This results in a weight ratio of Ti (4.08), which reflects the amount of material lost from the workpiece during the dry milling process. The corresponding wear mapping further illustrates the spatial distribution and severity of wear zones, reinforcing the influence of combined thermal, mechanical, and chemical interactions during the machining process of Ti-6Al-4 V alloy.

Unlike in dry machining, more severe wear mechanisms such as extensive notching, deep thermal cracking, or widespread adhesion were not prominently observed under Pulsated MQL conditions as shown in Fig. 4. This can be attributed to the partial lubrication effect of Pulsated MQL, which helps reduce friction at the tool-workpiece interface and thereby limits extreme temperature buildup. Lower temperatures reduce the likelihood of oxidation and thermal degradation of the tool surface, creating a more favorable environment for material transfer from the workpiece to the tool. As a result, a greater amount of Ti is deposited onto the cutting surface, leading to a Ti weight ratio of approximately 6, indicating the increased material transfer from the workpiece to the cutting insert. The presence of a fine oil mist forms a thin lubricating layer that minimizes direct metal-to-metal contact, suppressing the formation of built-up edge and reducing the likelihood of large-scale adhesive smearing [34]. Additionally, the moderated thermal environment prevents steep thermal gradients, which are typically responsible for the initiation of deep cracks and severe

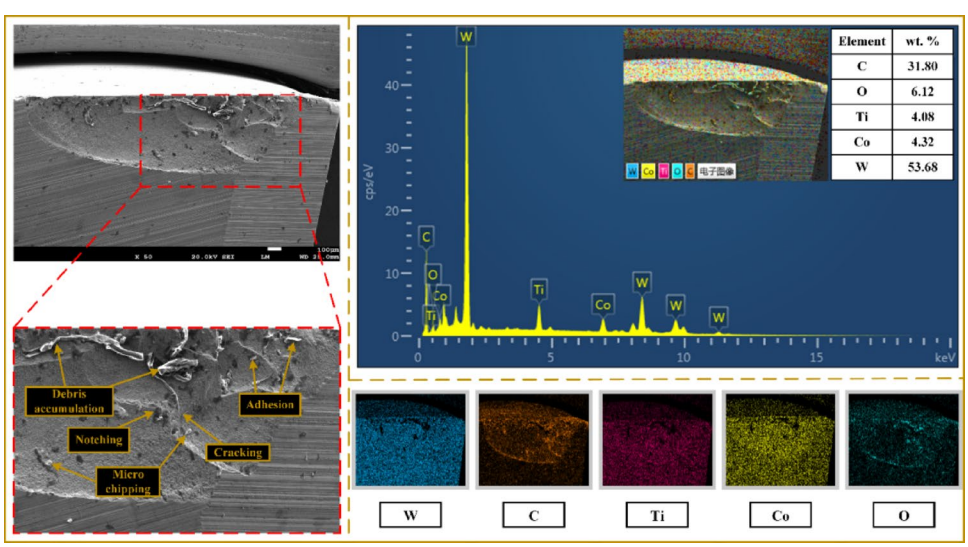

Fig. 3 Flank wear morphology and corresponding SEM-based tool mapping for Dry cutting environment

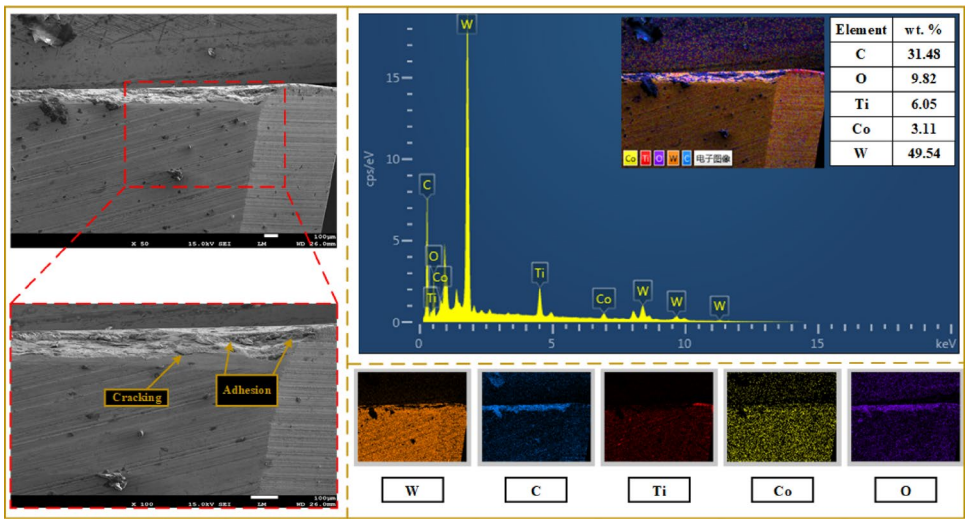

Fig. 4 Flank wear morphology and corresponding SEM-based tool mapping for pulsated MQL cutting environment

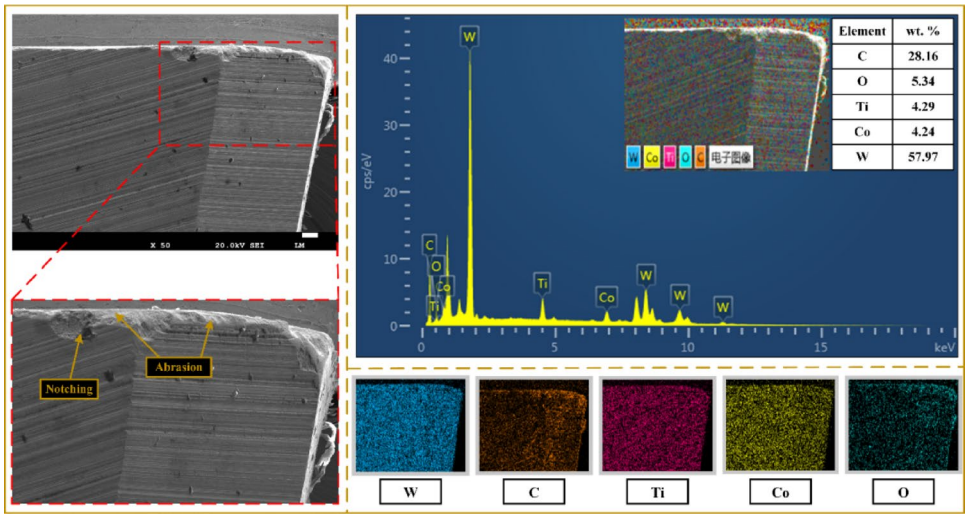

Fig. 5 Flank wear morphology and corresponding SEM-based tool mapping for Hybrid (pulsated MQL + dry ice) cutting environment

notching. As a result, while microcracking and microchipping were still present, the overall wear progression was less aggressive, indicating that Pulsated MQL offers a more balanced thermal-mechanical environment for tool longevity. The EDS analysis was repeated on the same region of the worn tool insert after proper cleaning. The revised spectrum now accurately represents the tool's material composition, as shown in Fig. 4, ensuring the validity of the compositional analysis.

Figure 5 shows the SEM of the flank face of the tool under Hybrid machining (pulsated MQL + dry ice). Among the three cutting environments evaluated, the Hybrid setup resulted in the minimum wear on the tool flank face. The observed wear mechanisms were limited to minor notching and slight abrasion, both of which were significantly less severe than the extensive damage observed under dry conditions and even more subdued compared to pulsated MQL.

The abrasive wear, though minimal, likely results from micro-interactions with hard particles or debris, but is effectively slowed by the cooling and lubricating synergy of the hybrid environment. The notching observed in this environment is attributed to localized thermal gradients between the tool and workpiece; however, the severity was notably reduced due to the effective heat dissipation provided by the dry ice [35]. Importantly, no chip adhesion or built-up material was detected on the tool surface. This is primarily due to the high-pressure CO_2 flow, which efficiently clears chips from the cutting zone, preventing reattachment or smearing. In this condition, CO_2 snow provides extreme cooling, which can have a dual effect on the machining process. While cooling is generally beneficial, but excessive cooling can lead to a brittle cutting zone, reducing the effectiveness of lubrication of MQL environment. As a result, the Ti weight ratio is approximately 4.29, indicating less material transfer compared to the MQL-only environment, where lubrication is more consistent and effective.

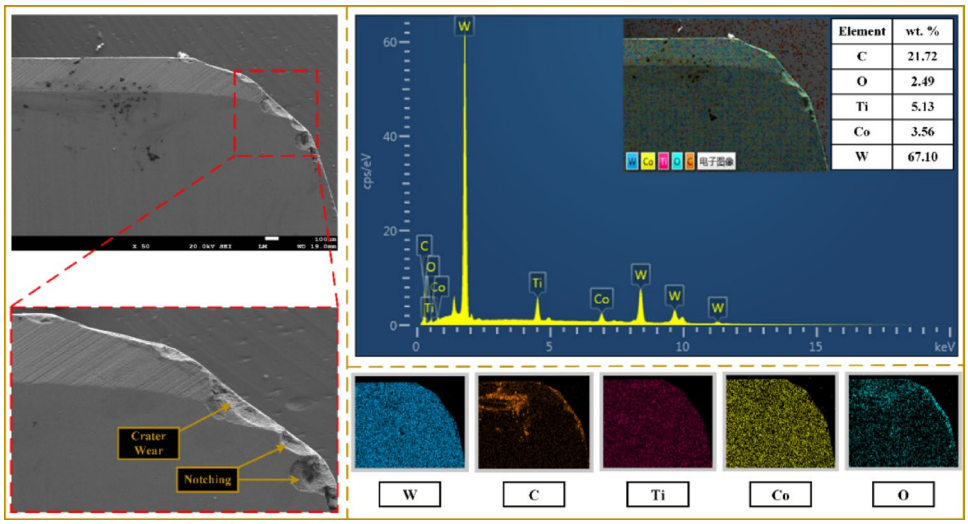

Fig. 6 Crater wear morphology and corresponding SEM-based tool mapping for dry cutting environment

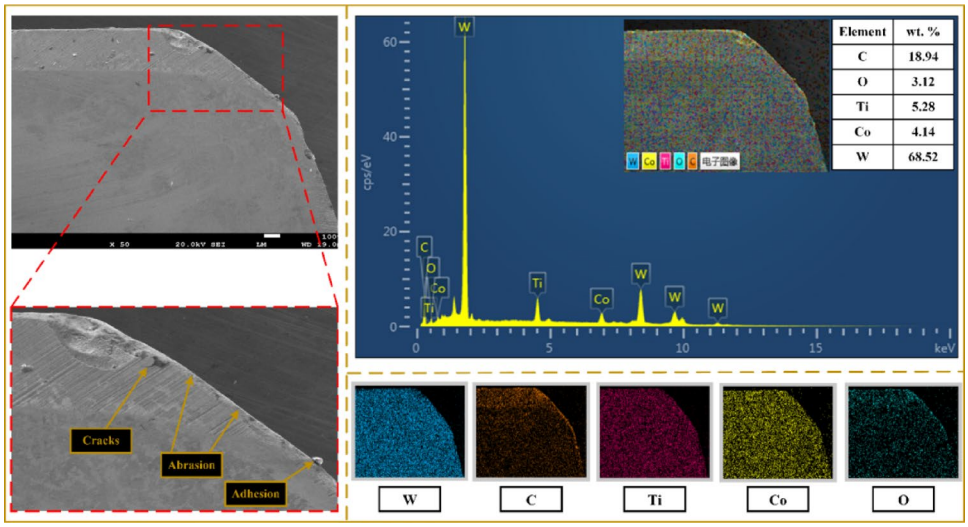

Fig. 7 Crater wear morphology and corresponding SEM-based tool mapping for pulsated MQL cutting environment

3.1.2 Tool Crater Wear

Crater wear on the rake face is primarily influenced by the high temperature and stress concentrations occurring in the primary shear zone, where material undergoes plastic deformation and interacts directly with the tool. The nature and severity of crater wear varied significantly across the three cooling environments due to differences in thermal control, lubrication, and chip-tool interaction. Under dry conditions, notching wear was the dominant form of crater damage as shown in Fig. 6. The absence of lubrication and cooling led to extreme temperatures at the tool-chip interface, accelerating tool-workpiece chemical affinity and promoting diffusion wear and thermal fatigue. Notching on the rake face is attributed to the cyclic mechanical loading imposed by chip flow over the tool surface, combined with localized thermal softening of the tool material [36]. The abrasive action of hard Ti-based chips, in the absence of a protective lubricant film, further intensifies wear at the depth-of-cut line where chip flow and stress concentration are highest.

In the pulsated MQL environment, minor cracks and abrasion were observed on the rake face as shown in Fig. 7. The application of a fine oil mist provided partial lubrication, which helped reduce friction and peak temperatures compared to dry cutting. This reduced the severity of notching and thermal wear mechanisms. However, the limited cooling effect of pulsated MQL could not fully prevent thermal cycling, leading to the initiation of microcracks through thermal fatigue. Abrasive wear also occurred due to incomplete separation between the chip and the rake face, especially under intermittent oil film breakdown. Despite this, pulsated MQL significantly suppressed adhesion wear and severe notching, thereby extending tool life compared to dry conditions.

In the Hybrid environment, only minor notching was observed, indicating the most effective control over crater wear mechanisms as shown on Fig. 8. The combined effect

of dry ice cooling and pulsated MQL lubrication maintained a low-temperature, low-friction cutting interface. The rapid sublimation of dry ice absorbed heat from the cutting zone, minimizing thermal gradients and reducing the likelihood of thermal fatigue and diffusion wear. Meanwhile, the lubricating film from pulsated MQL prevented metal-to-metal contact and minimized chip-tool adhesion [37]. As a result, cracking, adhesion, and abrasive damage were effectively suppressed, and crater wear was limited to slight mechanical notching at the chip impact area, likely caused by residual chip-tool interaction under high stress.

3.2 Machined Surface Roughness

Machined surface roughness R_a is typically defined as the arithmetic average of the deviations in surface height relative to a mean line, accounting for both peaks and valleys. It plays a critical role in determining the functional performance of machined components, as it influences friction, wear resistance, fatigue strength, and actual contact area. Among the various factors affecting surface roughness, the lubrication and cooling environment is particularly significant [38]. In the present study, three primary machining parameters were considered: cutting speed, feed rate, and lubrication condition, while other variables were kept constant. Surface roughness was measured using a profilometer with an evaluation length of 5.6 mm. Measurements were taken in two directions: along the feed and opposite to the feed direction. For each direction, three values were recorded, and the average of these values was subsequently calculated. The average, and standard deviation values are presented in the plots (Fig. 9) for each experimental condition, offering a comprehensive analysis of the surface finish under varying cooling and lubrication conditions.

The surface roughness of the machined Ti-6Al-4 V samples was measured at three locations along the cutting path beginning, middle, and end to ensure consistency and repeatability. Experimental results reveal that both cutting speed V_c and feed per tooth f_z significantly influence the surface finish. An increase in feed per tooth from 0.08 mm/z to 0.12 mm/z led to a noticeable rise in R_a values, whereas increasing the cutting speed resulted in a reduction in surface roughness. The improved surface finish at higher cutting speeds is due to lower machine excitation and a more stable chip process. On the other hand, further degradation of the surface finish at a higher feed rate might be caused by high temperatures at the tool–chip interface; these temperatures would cause poor evacuation of the chip and adhesion of the chip on the cutting edge and lead to worsening of the surface quality. At a feed rate of 0.08 mm/z, Hybrid (pulsated MQL + dry ice) cooling reduced surface roughness by 83.4% compared to Dry and 65.9% compared to pulsated MQL at 100 m/min, with similar reductions of 83.2% and 76.4% at 250 m/min. For 0.12 mm/z, the reductions were 78.2% (vs. dry) and 44.2% (vs. pulsated MQL) at 100 m/min, and 82.9% and 57.0%, respectively, at 250 m/min. Overall, Hybrid cooling consistently yielded the lowest roughness, confirming its effectiveness over dry and pulsated MQL conditions, especially at higher speeds. The results indicate that the copresent cooling and lubrication in the hybrid (pulsated MQL + dry ice) technique has a positive influence on the machined parts surface finish. In this combination, the dry ice serves as cooling medium to cool the cutting zone, while the pulsated MQL system ensures a sufficient supply of lubricant for the purpose of reducing the friction at the tool-workpiece contact area. Comparing the individual effects of dry ice and pulsated MQL methods, it is known that the high-pressure CO_2 snow is a more effective medium which helps to break the chip and prevent the chip sticking to the freshly machined surface. In addition, the vegetable oil from pulsated MQL system also builds up an anti-welding lubricant film during the machining process

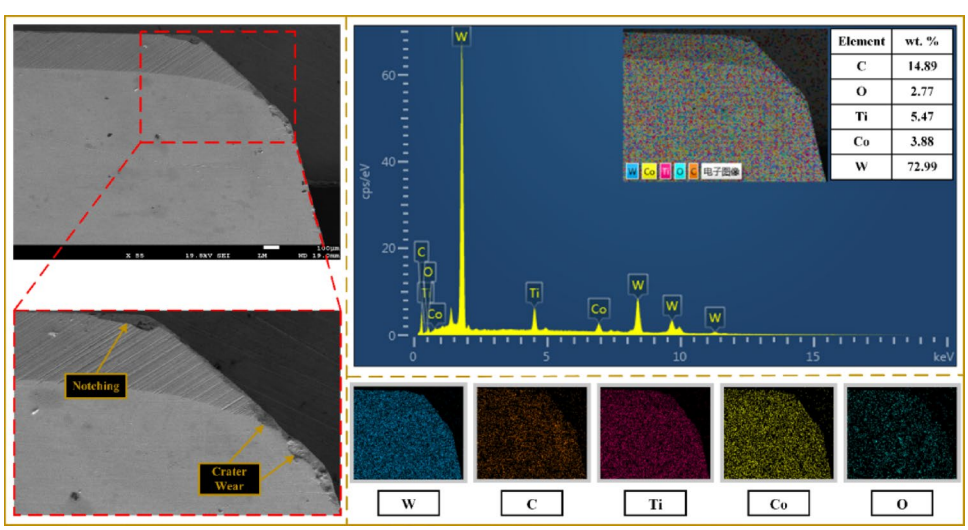

Fig. 8 Crater wear morphology and corresponding SEM-based tool mapping for Hybrid (Pulsated MQL + dry ice) cutting environment

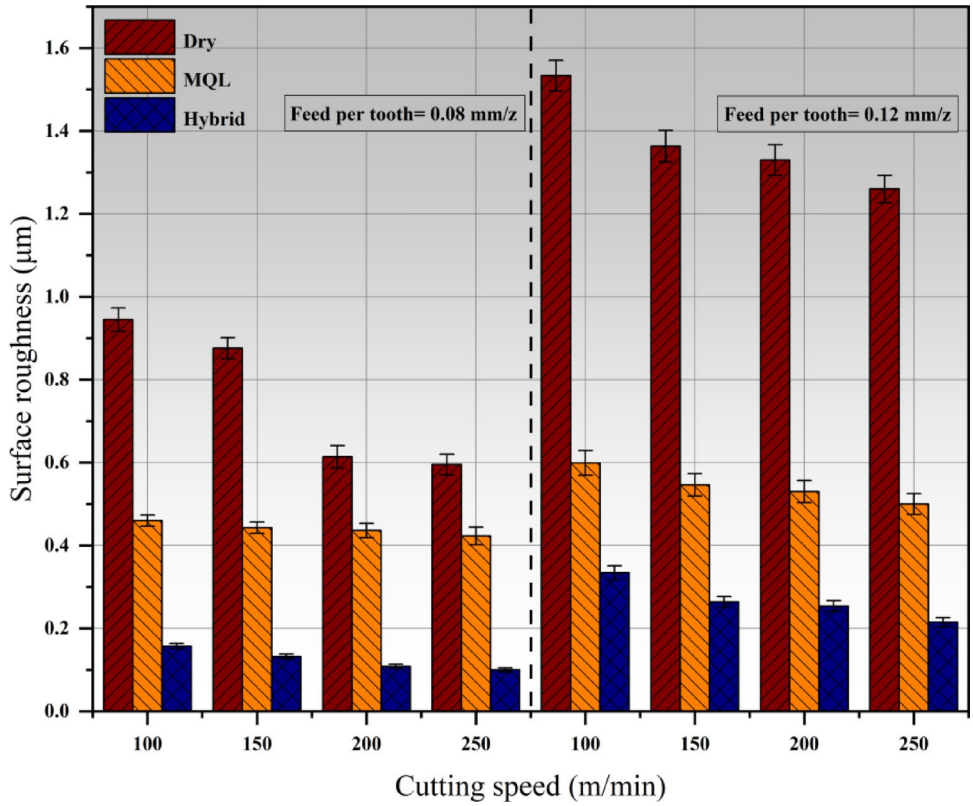

Fig. 9 Effect of cooling/lubrication on machined surface roughness under different cutting parameters

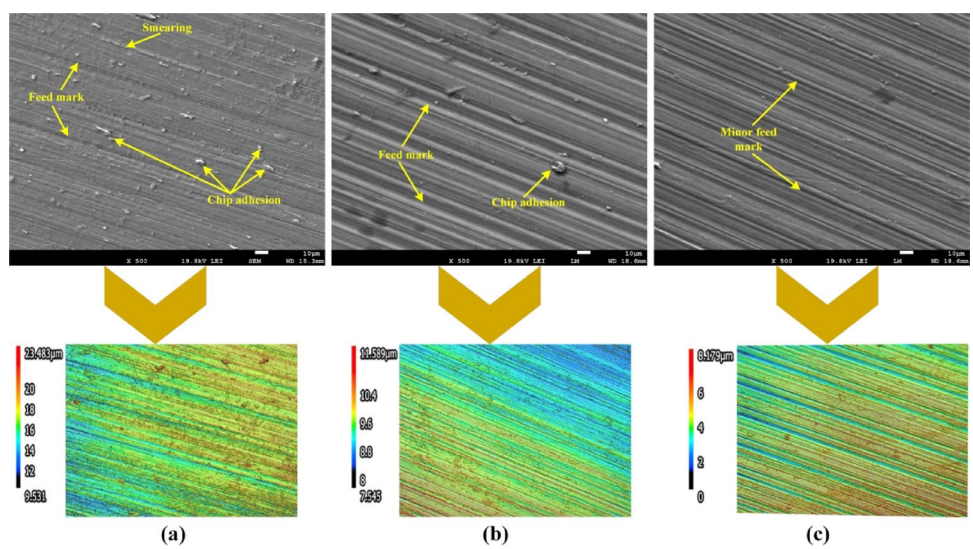

Fig. 10 SEM micrographs of machined surfaces at $V_c = 250$ m/min and $f_z = 0.12$ mm/rev under different cutting environments: (a) Dry, (b) Pulsated MQL, and (c) Hybrid (Pulsated MQL + dry ice)

and hence reduces the friction and the risk of chip welding [39]. Additionally, the fatty acid constituents in the vegetable oil chemically interact with the cutting material to form a metallic soap layer, which contributes to reduced frictional wear and improved surface finish. The kinetic energy of the dry ice particles also facilitates their rapid penetration into the cutting zone, aiding in heat dissipation and promoting efficient chip evacuation. Figure 10 presents the SEM analysis of the surface topography of Ti-6Al-4 V workpieces machined under three different cutting environments: dry, pulsated MQL, and hybrid (pulsated MQL + dry ice). The surface morphology revealed that the hybrid cooling environment produced the most favorable surface quality, characterized by minimal feed marks, reduced material damage, and a cleaner surface finish. In contrast, surfaces machined under dry conditions exhibited pronounced feed marks, extensive smearing, and visible chip adhesion, as shown in Fig. 10(a). This degradation in surface quality is attributed to the poor thermal conductivity of Ti-6Al-4 V, which limits heat dissipation through the workpiece and leads to elevated

surface temperatures. The increased thermal load softens the material, resulting in plastic deformation and smearing, where the material spreads across the surface without being effectively cut [40].

Under pulsated MQL conditions, the presence of lubrication reduced frictional effects and partially suppressed smearing; however, chip adhesion remained evident due to limited chip evacuation. Chip re-deposition was observed across all environments; nevertheless, the hybrid cooling strategy significantly minimized this phenomenon. The superior performance of the hybrid system can be attributed to the combined benefits of high-pressure CO_2, which enhanced chip breakability and facilitated chip evacuation, and the effective lubrication from pulsated MQL, which reduced tool-workpiece friction [27]. This synergy between cooling and lubrication prevented chip welding, promoted chip curling, and contributed to improved surface integrity. Overall, the SEM images confirmed that the Hybrid approach effectively suppressed smearing and plastic deformation, resulting in a significantly enhanced surface finish compared to Dry and pulsated MQL conditions.

3.3 X-ray Diffraction (XRD) Tests of Machined Surface

The XRD analysis of Ti-6Al-4 V workpieces machined under varying cutting environments dry, pulsated MQL, and hybrid (pulsated MQL + dry ice) revealed distinct differences in surface phase evolution and crystallographic integrity as depicted in Fig. 11. In all three conditions, prominent peaks were observed at 2θ values around 35°, 38°, and 40°, corresponding to the base material constituent's titanium (Ti) and aluminum (Al), which form the α and β phases of the Ti-6Al-4 V alloy. However, the intensity and

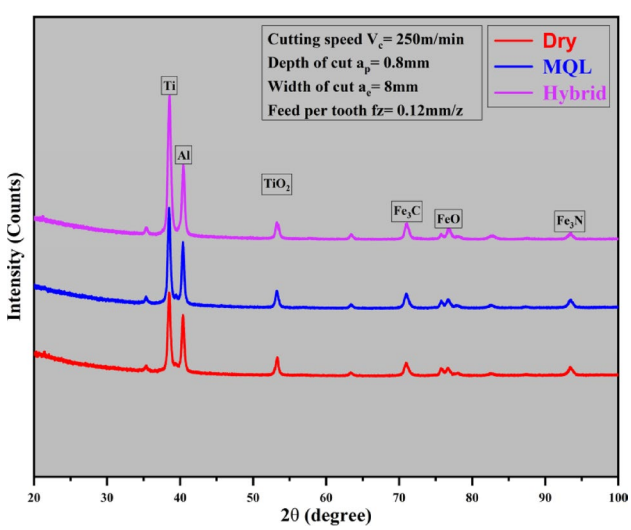

Fig. 11 Effects of the cutting environments on the XRD spectra of TC4 alloy machined under high-speed face milling

sharpness of these peaks were highest in the hybrid condition, indicating a more preserved and undisturbed crystal structure, whereas the peaks under dry and pulsated MQL conditions were relatively broadened, suggesting increased micro strain and possible lattice distortion due to thermal and mechanical effects during machining [41].

Additionally, secondary peaks detected at higher angles, particularly between $2\theta = 50°–80°$, correspond to oxide and carbide phases such as TiO_2 (55°), Fe_3C (65°), FeO (70°), and Fe_3N (75°). These peaks were most intense under dry cutting conditions, slightly less pronounced under pulsated MQL, and significantly reduced in the hybrid condition [42]. This trend indicates that the dry environment promoted higher surface oxidation and thermomechanical interactions due to elevated cutting temperatures and insufficient lubrication. Pulsated MQL partially mitigated these effects through limited lubrication, but still allowed for some thermal degradation. In contrast, the hybrid environment, utilizing both pulsated MQL and dry ice, effectively suppressed the formation of these thermal-induced phases by enhancing heat dissipation and reducing the temperature at the tool-workpiece interface.

The decreased intensity of TiO_2, Fe_3C, and Fe_3N peaks under the hybrid condition implies reduced surface oxidation, carbide formation, and nitrogen diffusion phenomena typically accelerated at elevated temperatures [43]. Simultaneously, the enhancement of Ti and Al peak intensities supports the preservation of base alloy characteristics. These findings clearly demonstrate that the hybrid cooling approach offers superior protection of surface crystallinity and microstructure during high-speed face milling of Ti-6Al-4 V, making it a more effective strategy compared to dry and conventional pulsated MQL machining environments.

3.4 Cutting Forces

In milling operations, cutting forces are generated in three orthogonal directions to deform the material and facilitate chip formation. These forces play a critical role in the design of machine tool structures, estimation of power requirements, prediction of tool life, and optimization of machining productivity. Among them, the tangential cutting force is the most dominant, accounting for approximately 70% of the total force, as it primarily resists the rotational motion of the tool. The feed force, representing about 20%, acts in the direction of tool advancement, while the radial force, comprising the remaining 10%, acts perpendicular to the feed direction and tends to deflect the tool away from the workpiece [44].

For each experimental condition, the cutting forces were measured in three repetitions. In each repetition, the peak cutting force was recorded as the maximum force observed

during the milling process. These peak values were then averaged across the three repetitions to obtain a mean cutting force for each experimental condition. To represent the variability of the data, standard deviation was calculated from the three repetitions and error bars were included in the corresponding plots. Figure 12 shows the plot of axial cutting force f_z with changing cutting speed and feed/tooth. Cutting forces were higher with the larger feed per tooth as more material was engaged per revolution, inducing higher cutting load. However, the forces reduced with increasing cutting speed, because of shortening in the tool-workpiece contact time. This tendency was repeatedly observed in the dry, pulsated MQL, and hybrid coolant conditions. Cutting force was lowest for hybrid (pulsated MQL + dry ice) cooling and then for pulsated MQL, and it was highest for dry cutting. At low cutting speed of 100 m/min and a feed per tooth of 0.08 mm/z, combined pulsated MQL and dry ice ambient has caused that axial cutting force is around 44% less than the dry condition and 19% less than pulsated MQL. At a higher speed of 250 m/min, hybrid reduced the forces by about 44% and 32% over Dry and pulsated MQL, respectively. At a feed rate of 0.12 mm/z, similar reductions were observed at 100 m/min, for which hybrid reduced axial force by 41% over dry and 34% over pulsated MQL. Comparatively, while at the cutting speed of 250 m/min, the reductions were changed to 40% and 37%, respectively.

The lower cutting forces developed in the hybrid environment was due to the improved chip removal and significant reduction in friction at tool-chip interface, coupled with the combined action of effective lubrication by pulsated MQL and intense cooling by CO_2. This cooling-lubrication combination effectively suppresses thermal softening of the workpiece material. Although thermal softening may appear advantageous, it often leads to increased cutting forces due to material adhesion at elevated temperatures. The softened material tends to weld onto the cutting edge, distorting the tool geometry and thereby increasing cutting resistance [45]. Under pulsated MQL conditions, cutting forces are generally lower than in dry machining due to the presence of a fine mist of lubricant that reduces friction at the tool-chip and tool-workpiece interfaces. The lubricant forms a thin film that minimizes metal-to-metal contact, thereby decreasing cutting resistance. Although pulsated MQL provides limited cooling compared to hybrid systems, it is often sufficient to reduce the temperature rise at the cutting zone to a moderate extent. This helps in preventing severe thermal softening, built-up edge formation, and excessive tool wear. However, due to the restricted cooling capacity of pulsated MQL, it may not fully suppress temperature-induced effects such as ductility enhancement or oxidation, which can still contribute to moderate cutting forces-higher than hybrid, but lower than dry conditions [46]. In Dry cutting conditions, the absence of lubrication and cooling results in elevated cutting zone temperatures, which significantly increase friction and promote material adhesion at the tool-chip interface. The associated thermal softening

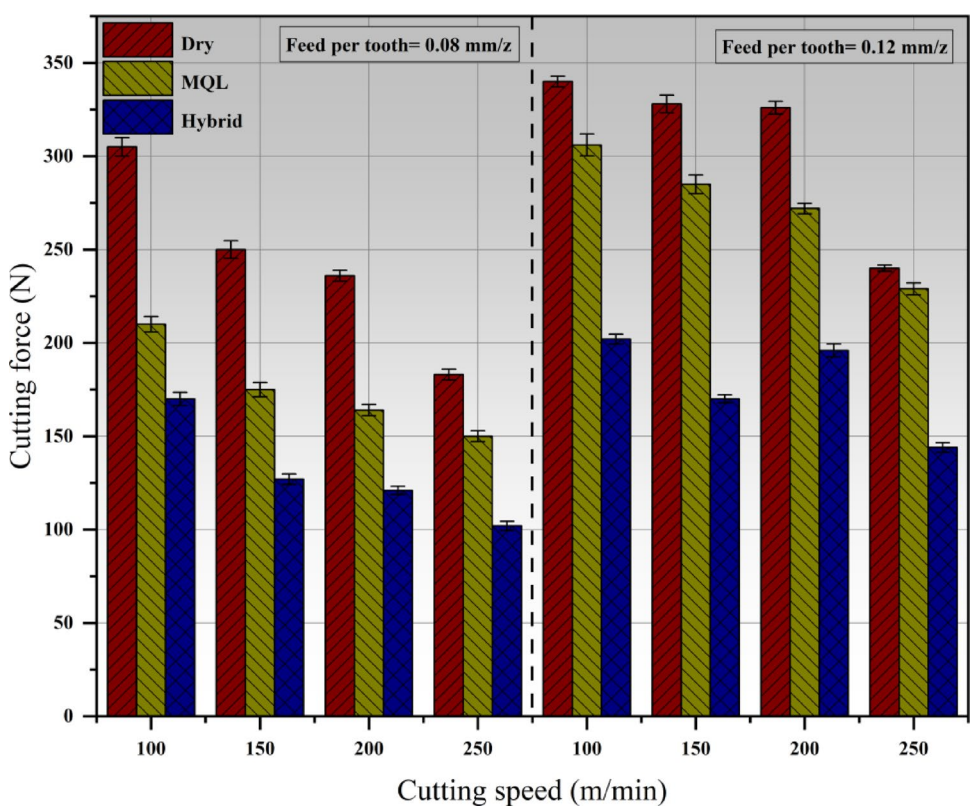

Fig. 12 Influence of axial cutting force under different cutting environments (dry, pulsated MQL, hybrid) at varying cutting speeds and feed rates

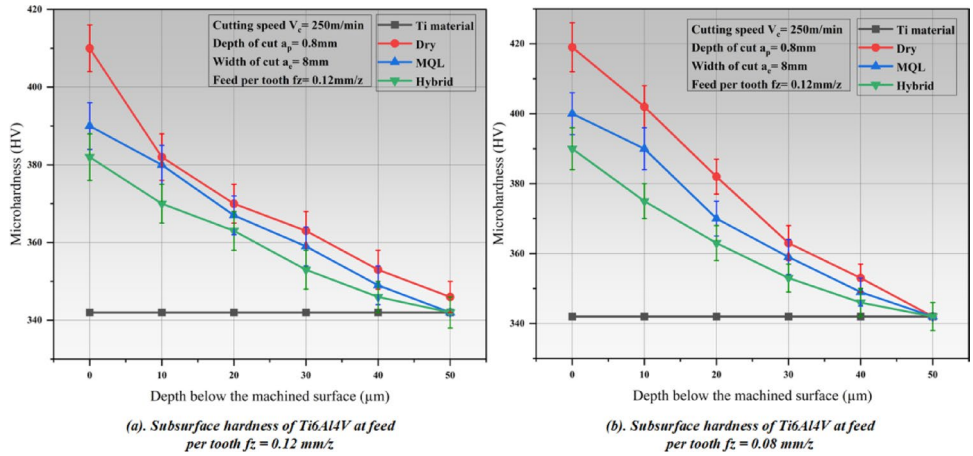

Fig. 13 Subsurface microhardness distribution of Ti-6Al-4 V under dry, pulsated MQL, and hybrid (pulsated MQL + dry ice) machining environments

enhances the ductility of the workpiece material, leading to larger plastic deformation zones and increased cutting energy requirements. Also, elevated temperatures can cause oxidation and localized surface hardening, which results in larger forces of the tool-workpiece interaction. Furthermore, higher contact stresses and machining instability due to severe thermal effects, rapid tool wear and thermal expansion bring about higher cutting forces in dry conditions [47]. Lower cutting forces resulted in direct improvements in machine tools productivity, reduced energy requirement and increased tool life, which improved the overall sustainability of the machining process.

3.5 Surface micro-hardness

Surface and subsurface microhardness measurements are also very important in understanding how much plastic deformation was caused by machining, and how deep this effect penetrated into the material. This investigation, however, also useful to explore thermally induced effects, for example phase change due to increased temperature and changes in cooling conditions in the cutting zone. Such analyses are necessary in order to evaluate the machining condition and surface integrity. Figure 13 illustrates the surface and subsurface microhardness distributions of Ti-6Al-4 V alloy machined under Dry, pulsated MQL, and Hybrid (pulsated MQL + dry ice CO_2) cooling environments at varying feed rates.

The initial (bulk) hardness of the Ti-6Al-4 V workpiece was approximately 342 HV. At 0 μm depth, hybrid cooling reduced surface hardness by 6.9% vs. dry and 2.5% vs. pulsated MQL at 0.12 mm/z feed, while by 6.8% vs. dry and 2.1% vs. pulsated MQL at 0.08 mm/z, confirming its effectiveness in suppressing surface hardening. The results clearly indicate that dry machining consistently produced the highest surface and subsurface hardness, followed by pulsated MQL, while the hybrid environment yielded the lowest hardness values at all measured depths. This trend was observed at both feed rates (0.08 mm/z and 0.12 mm/z), with a notable decrease in hardness with increasing feed rate across all environments [35]. This increase was more pronounced at the lower feed rate (0.08 mm/z), indicating the combined effect of severe plastic deformation and thermal hardening due to the absence of cooling and lubrication. The plastic flow rate allowed primary shear zone heat generation results in variation of hardening effects. Moreover, the reduction in microhardness at higher feed rate is attributed to higher material removal rates at increased feed, which produce more heat and reduce the severity of localized deformation per unit volume, thereby diminishing the hardening effect near the surface [48]. The elevated hardness under dry conditions can be attributed to intense plastic deformation and higher cutting temperatures caused by the lack of cooling and lubrication. These conditions promote severe strain hardening and may induce quenching-like effects due to rapid self-cooling by ambient air, leading to localized microstructural hardening near the surface.

In contrast, the pulsated MQL environment introduces a limited quantity of lubricant, which provides partial reduction in friction and moderate control of the cutting temperature. The result is a less severe thermal and mechanical load, leading to reduced plastic deformation in the surface and subsurface layers. Consequently, strain hardening occurs to a lesser extent than in dry cutting. The lubrication effect also limits the severity of tool-workpiece adhesion and cutting-edge wear, which further stabilizes the cutting process and reduces subsurface microstructural disturbance [49]. As a result, the hardness values observed under pulsated MQL fall between those of dry and hybrid environments.

The hybrid pulsated MQL and dry ice environment, which integrates the lubricating effect of pulsated MQL with the dry ice CO_2, provides superior thermal mitigation during machining. At elevated cutting speeds, where heat generation is significant, the sublimation of dry ice in the hybrid mist effectively absorbs and removes heat from the cutting zone. The high-velocity CO_2 particles enhance penetration

into the tool–workpiece interface, allowing the cooling medium to reach closer to the shear zone. This synergistic action of lubrication and dry ice cooling reduces both frictional and cutting heat, thereby lowering the thermal load on the workpiece [50]. The resulting temperature reduction prevents excessive thermal softening and significantly suppresses plastic deformation. Moreover, the minimized temperature gradients in the Hybrid environment help reduce the formation of residual stresses and inhibit undesirable phase transformations, leading to a more stable and uniform hardness profile that closely resembles the bulk material.

4 Conclusions

In this study, milling experiments were conducted on Ti-6Al-4 V alloy under three distinct ecological cooling and lubrication environments including dry, pulsated MQL, and hybrid (i.e., pulsated MQL combined with dry ice) scenarios. The performance of each environment was assessed by a holistic study of tool wear, machined surface roughness and topography, subsurface microhardness and cutting forces. Based on the experimental findings, the key conclusions are summarized as follows:

- Tool wear was significantly reduced under the hybrid pulsated MQL and dry ice environment compared to pulsated MQL and dry conditions. Flank wear was primarily governed by chipping, notching, and cracking, with adhesion being the dominant mechanism under dry cutting. The hybrid setup effectively suppressed adhesion and thermal-related damage. SEM analysis of the rake face revealed only minor notching under hybrid condition, indicating improved wear resistance and tool stability.
- The hybrid pulsated MQL and dry ice environment yielded the lowest surface roughness across all cutting conditions. Compared to dry and pulsated MQL, it reduced R_a by up to 83.4% and 76.4% at 0.08 mm/z, and by 82.9% and 57.0% at 0.12 mm/z, respectively. These improvements were consistent with surface topography observations, confirming enhanced surface quality under Hybrid condition.
- XRD analysis revealed that the hybrid cooling environment (i.e., pulsated MQL and dry ice) minimized the formation of secondary phases and best preserved the crystalline structure of the Ti-6Al-4 V surface, indicating superior control over thermal effects during machining as compared to pulsated MQL and dry conditions.
- The hybrid pulsated MQL and dry ice environment significantly reduced axial cutting forces across all cutting conditions. At 0.08 mm/z, it achieved reductions of up to 44% compared to dry and 32% compared to pulsated MQL, while at 0.12 mm/z, the maximum reductions were 41% and 37%, respectively. These results highlight the hybrid system's effectiveness in minimizing cutting resistance through enhanced cooling and lubrication.
- The hybrid pulsated MQL and dry ice environment exhibited the lowest surface microhardness values among the tested conditions, reflecting its ability to suppress thermally induced strain hardening. At 0 μm depth, surface hardness was reduced by 6.9% compared to dry and 2.5% compared to pulsated MQL at a feed rate of 0.12 mm/z, and by 6.8% and 2.1%, respectively, at 0.08 mm/z. These findings confirm the hybrid system's effectiveness in maintaining surface integrity through improved thermal control.

Acknowledgements The authors would like to acknowledge the financial support from the National Natural Science Foundation of China [52522509, 52175420] and the National Key Research and Development Program of China (2023YFB3407300). This work was also supported by grants from the Natural Science Foundation of Shandong Province (ZR2024ZD43), Taishan Scholar Program of Shandong Province [tsqn202103015], and the Self-developed Instrument and Equipment Project of Shandong University (zy20240303).

Data Availability Not applicable.

Code Availability Not applicable.

Declarations

Conflict of interest The authors declare no conflict of interest.

Ethics Approval Not applicable.

Consent To Participate Not applicable.

Consent for publication Not applicable

References

1. Lindvall, R., Gayubo, J. C., Gutnichenko, O., Auzenat, F., M'Saoubi, R., & Bushlya, V. (2025). Performance and wear mechanisms of TiAlN-NbN coated cemented carbide in milling Ti6Al4V with different cooling and lubrication approaches. *Wear*, *205846*. https://doi.org/10.1016/j.wear.2025.205846
2. Wang, B., Liu, Z. Q., Cai, Y. K., Luo, X. C., Ma, H. F., Song, Q. H., & Xiong, Z. H. (2023). Advancements in material removal mechanism and surface integrity of high speed metal cutting: A review. *International Journal of Machine Tools and Manufacture, 166*, Article 103744. https://doi.org/10.1016/j.ijmachtools.2021.103744
3. Khan, M. A., Wang, B., Khan, A. M., Liu, Z., Cai, Y., & Song, Q. (2024). Integrating textured tools and diverse lubricooling approaches in machining performance of Ti and Ni-base alloys – A comprehensive review. *Machining Science and Technology, 28*(3), 323–391. https://doi.org/10.1080/10910344.2024.2324828

4. Marinosci, V. M., Chu, L., Grouve, W. J. B., Wijskamp, S., Akkerman, R., & de Rooij, M. B. (2022). Characterization of the water–titanium interaction and its effect on the adhesion of titanium-C/PEKK joints. *Composites Part A, Applied Science and Manufacturing, 162*, 107107. https://doi.org/10.1016/j.compositesa.2022.107107
5. Kaynak, Y., Gharibi, A., Yılmaz, U., Köklü, U., & Aslantaş, K. (2018). A comparison of flood cooling, minimum quantity lubrication and high pressure coolant on machining and surface integrity of titanium Ti-5553 alloy. *Journal of Manufacturing Processes*, *34*, 503–512. https://doi.org/10.1016/j.jmapro.2018.06.003
6. Padhan, S., Mishra, S., Sahu, S. K., Das, A., Das, S. R., & Mohapatra, S. (2025). Enhancing machining performance and sustainability: A comprehensive review of minimum quantity lubrication. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, *47*(7), 316. https://doi.org/10.1007/s40430-025-05626-6
7. Sankaranarayanan, R., & Krolczyk, G. M. (2021). A comprehensive review on research developments of vegetable-oil based cutting fluids for sustainable machining challenges. *Journal of Manufacturing Processes*, *67*, 286–313. https://doi.org/10.1016/j.jmapro.2021.05.002
8. Zhao, J., Liu, Z., Wang, B., Song, Q., Cai, Y., Khan, A. M., Wan, Y., & Ren, X. (2023). A comprehensive review of generating, monitoring, evaluating, and controlling particle emissions during machining process. *Journal of Manufacturing Systems*, *70*, 395–416. https://doi.org/10.1016/j.jmsy.2023.08.007
9. Ji, C., Sheng, R., Wu, H., Zhou, Z., Yan, X., Dong, L., & Li, C. (2024). Bibliometric analysis and research trends in minimum quantity lubrication for reducing cutting forces. *The International Journal of Advanced Manufacturing Technology*, *135*(11), 4995–5033. https://doi.org/10.1007/s00170-024-14793-0
10. Sharma, A. K., Tiwari, A. K., & Dixit, A. R. (2016). Effects of minimum quantity lubrication (MQL) in machining processes using conventional and nanofluid based cutting fluids: A comprehensive review. *Journal of Cleaner Production*, *127*, 1–18. https://doi.org/10.1016/j.jclepro.2016.03.146
11. Said, Z., Gupta, M., Hegab, H., Arora, N., Khan, A. M., Jamil, M., & Bellos, E. (2019). A comprehensive review on minimum quantity lubrication (MQL) in machining processes using nanocutting fluids. *The International Journal of Advanced Manufacturing Technology*, *105*, 2057–2086.
12. Das, A., Das, S. R., Patel, S. K., & Biswal, B. B. (2020). Effect of MQL and nanofluid on the machinability aspects of hardened alloy steel. *Machining Science and Technology*, *24*(2), 291–320. https://doi.org/10.1080/10910344.2019.1669167
13. Sen, B., Mia, M., Krolczyk, G. M., Mandal, U. K., & Mondal, S. P. (2021). Eco-friendly cutting fluids in minimum quantity lubrication assisted machining: A review on the perception of sustainable manufacturing. *International Journal of Precision Engineering and Manufacturing-Green Technology, 8*(1), 249–280. https://doi.org/10.1007/s40684-019-00158-6
14. Iyappan, S. K., & Ghosh, A. (2020). Small quantity lubrication assisted end milling of aluminium using sunflower oil. *International Journal of Precision Engineering and Manufacturing-Green Technology*, *7*(2), 337–345. https://doi.org/10.1007/s40684-019-00081-w
15. Vazquez, E., Gomar, J., Ciurana, J., & Rodríguez, C. A. (2015). Analyzing effects of cooling and lubrication conditions in micromilling of Ti6Al4V. *Journal of Cleaner Production*, *87*, 906–913. https://doi.org/10.1016/j.jclepro.2014.10.016
16. Hadad, M., & Sadeghi, B. (2013). Minimum quantity lubrication-MQL turning of AISI 4140 steel alloy. *Journal of Cleaner Production*, *54*, 332–343. https://doi.org/10.1016/j.jclepro.2013.05.011
17. Sarikaya, M., Gupta, M. K., Tomaz, I., Danish, M., Mia, M., Rubaiee, S., Jamil, M., Pimenov, D. Y., & Khanna, N. (2021). Cooling techniques to improve the machinability and sustainability of light-weight alloys: A state-of-the-art review. *Journal of Manufacturing Processes*, *62*, 179–201. https://doi.org/10.1016/j.jmapro.2020.12.013
18. He, T., Liu, N., Xia, H., Wu, L., Zhang, Y., Li, D., & Chen, Y. (2023). Progress and trend of minimum quantity lubrication (MQL): A comprehensive review. *Journal of Cleaner Production*, *386*, 135809. https://doi.org/10.1016/j.jclepro.2022.135809
19. Zhou, S., Wang, D., Wu, S., Gu, G., Dong, G., An, Q., Guo, H., & Li, C. (2024). Minimum quantity lubrication machining nickel base alloy: A comprehensive review. *The International Journal of Advanced Manufacturing Technology*, *131*(5), 2407–2445. https://doi.org/10.1007/s00170-023-11721-6
20. Nath, C., Kapoor, S. G., DeVor, R. E., Srivastava, A. K., & Iverson, J. (2012). Design and evaluation of an atomization-based cutting fluid spray system in turning of titanium alloy. *Journal of Manufacturing Processes*, *14*(4), 452–459. https://doi.org/10.1016/j.jmapro.2012.09.002
21. Nath, C., Kapoor, S. G., & Srivastava, A. K. (2017). Finish turning of Ti-6Al-4V with the atomization-based cutting fluid (ACF) spray system. *Journal of Manufacturing Processes*, *28*, 464–471. https://doi.org/10.1016/j.jmapro.2017.04.013
22. Roy, S., Kumar, R., Sahoo, A. K., & Panda, A. (2020). Cutting tool failure and surface finish analysis in pulsating MQL-assisted hard turning. *Journal of Failure Analysis and Prevention, 20*(4), 1274–1291. https://doi.org/10.1007/s11668-020-00940-8
23. Jebaraj, M., & Pradeep Kumar, M. (2019). Effect of cryogenic CO2 and LN2 coolants in milling of aluminum alloy. *Materials and Manufacturing Processes*, *34*(5), 511–520. https://doi.org/10.1080/10426914.2018.1532591
24. Jebaraj, M., Kumar, P., & Anburaj, R. (2020). Effect of LN_2 and CO_2 coolants in milling of 55NiCrMoV7 steel. *Journal of Manufacturing Processes, 53*, 318–327. https://doi.org/10.1016/j.jmapro.2020.02.040
25. Zhao, W., Ren, F., Iqbal, A., Gong, L., He, N., & Xu, Q. (2020). Effect of liquid nitrogen cooling on surface integrity in cryogenic milling of Ti-6Al-4 V titanium alloy. *The International Journal of Advanced Manufacturing Technology*, *106*(3), 1497–1508. https://doi.org/10.1007/s00170-019-04721-y
26. Sivaiah, P., & Chakradhar, D. (2018). Effect of cryogenic coolant on turning performance characteristics during machining of 17 – 4 PH stainless steel: A comparison with MQL, wet, dry machining. *CIRP Journal of Manufacturing Science and Technology*, *21*, 86–96.
27. Tu, L., Chen, J., An, Q., Ming, W., Xu, J., Chen, M., Lin, L., & Yang, Z. (2021). Machinability improvement of compacted graphite irons in milling process with supercritical CO2-based MQL. *Journal of Manufacturing Processes*, *68*, 154–168. https://doi.org/10.1016/j.jmapro.2021.05.044
28. Wika, K. K., Litwa, P., & Hitchens, C. (2019). Impact of supercritical carbon dioxide cooling with minimum quantity lubrication on tool wear and surface integrity in the milling of AISI 304L stainless steel. *Wear*, *426–427*, 1691–1701. https://doi.org/10.1016/j.wear.2019.01.103
29. Jamil, M., Khan, A. M., Gupta, M. K., Mia, M., He, N., Li, L., & Sivalingam, V. (2020). Influence of CO2-snow and subzero MQL on thermal aspects in the machining of Ti-6Al-4V. *Applied Thermal Engineering*, *177*, 115480. https://doi.org/10.1016/j.applthermaleng.2020.115480
30. Rahman, M., Wang, Z. G., & Wong, Y. S. (2006). A review on high-speed machining of titanium alloys. *JSME International Journal Series C, 49*(1), 11–20. https://doi.org/10.1299/jsmec.49.11

31. Deng, B., He, Q., DePaiva, J. M., & Veldhuis, S. C. (2022). A novel approach to cutting tool edge design based on initial wear stage. *Journal of Materials Processing Technology*, *304*, 117561. https://doi.org/10.1016/j.jmatprotec.2022.117561
32. Hourmand, M., Sarhan, A. A. D., Sayuti, M., & Hamdi, M. (2021). A comprehensive review on machining of titanium alloys. *Arabian Journal for Science and Engineering*, *46*, 7087–7123.
33. George, A., Dhanish, P. B., Mathew, J., & Kuriachen, B. (2024). Wear characteristics of WS2 nanoparticle-assisted micro-EDM textured carbide inserts during machining of Ti6Al4V. *Tribology International*, *195*, 109640. https://doi.org/10.1016/j.triboint.2024.109640
34. Li, D., Zhang, T., Zheng, T., Zhao, N., & Li, Z. (2024). A comprehensive review of minimum quantity lubrication (MQL) machining technology and cutting performance. *The International Journal of Advanced Manufacturing Technology*, *133*(5), 2681–2707. https://doi.org/10.1007/s00170-024-13902-3
35. Jamil, M., He, N., Zhao, W., Khan, A. M., Xiang, H., Gupta, M. K., & Iqbal, A. (2022). A novel low-pressure hybrid dry ice blasting system for improving the tribological and machining characteristics of AISI-52100 tool steel. *Journal of Manufacturing Processes*, *80*, 152–160.
36. Palaniappan, K., Sundararaman, M., Murthy, H., Jeyaraam, R., & Rao, B. C. (2022). Influence of workpiece texture and strain hardening on chip formation during machining of Ti–6Al–4V alloy. *International Journal of Machine Tools and Manufacture*, *173*, 103849. https://doi.org/10.1016/j.ijmachtools.2021.103849
37. Halim, N. H. A., Haron, C. H. C., Ghani, J. A., & Azhar, M. F. (2019). Tool wear and chip morphology in high-speed milling of hardened inconel 718 under dry and cryogenic CO2 conditions. *Wear*, *426*, 1683–1690.
38. Cheng, Y., Wang, Y., Lin, J., Xu, S., & Zhang, P. (2023). Research status of the influence of machining processes and surface modification technology on the surface integrity of bearing steel materials. *The International Journal of Advanced Manufacturing Technology*, *125*(7), 2897–2923. https://doi.org/10.1007/s00170-023-10960-x
39. Danish, M., Gupta, M. K., Rubaiee, S., Ahmed, A., & Korkmaz, M. E. (2021). Influence of hybrid Cryo-MQL lubri-cooling strategy on the machining and tribological characteristics of inconel 718. *Tribology International*, *163*, 107178.
40. Zhang, D., Zhang, X. M., Nie, G. C., Yang, Z. Y., & Ding, H. (2021). Characterization of material strain and thermal softening effects in the cutting process. *International Journal of Machine Tools and Manufacture*, *160*, 103672. https://doi.org/10.1016/j.ijmachtools.2020.103672
41. Yaqoob, S., Ghani, J. A., Juri, A. Z., Muhamad, S. S., Haron, C. H. C., & Jouini, N. (2024). A review of sustainable hybrid lubrication (Cryo-MQL) techniques in machining processes. *The International Journal of Advanced Manufacturing Technology*, *131*(1), 151–169. https://doi.org/10.1007/s00170-024-13135-4
42. Nawaz, M. A., Saif, M., Li, M., Song, G., Zihao, W., Chen, C., & Liu, D. (2022). Elucidating the synergistic fabrication of dual embedded (χ-Fe5C2 + θ-Fe3C) carbide nanocomposites in Na-FeCa@AC/HZSM-5 integrated catalyst for syngas conversion to aromatics. *Fuel, 324*, 124390. https://doi.org/10.1016/j.fuel.2022.124390
43. Hou, L., Yang, W., Xu, X., Deng, B., Chen, Z., Wang, S., Tian, J., Yang, F., & Li, Y. (2019). In-situ activation endows the integrated Fe3C/Fe@nitrogen-doped carbon hybrids with enhanced pseudocapacitance for electrochemical energy storage. *Chemical Engineering Journal*, *375*, 122061. https://doi.org/10.1016/j.cej.2019.122061
44. Bashir, M., Al, Mia, M., & Dhar, N. R. (2018). Investigations on surface milling of hardened AISI 4140 steel with pulse jet MQL applicator. *Journal of the Institution of Engineers (India): Series C*, *99*(3), 301–314. https://doi.org/10.1007/s40032-016-0277-2
45. Li, B., Zhang, S., Du, J., & Sun, Y. (2022). State-of-the-art in cutting performance and surface integrity considering tool edge micro-geometry in metal cutting process. *Journal of Manufacturing Processes*, *77*, 380–411. https://doi.org/10.1016/j.jmapro.2022.03.037
46. Khosravi, J., Azarhoushang, B., Barmouz, M., Bösinger, R., & Zahedi, A. (2022). High-speed milling of Ti6Al4V under a supercritical CO2 + MQL hybrid cooling system. *Journal of Manufacturing Processes*, *82*, 1–14. https://doi.org/10.1016/j.jmapro.2022.07.061
47. Shi, L., Yin, Y., Wang, C., Wang, T., Yang, L., Wang, Q., An, Q., Ming, W., & Chen, M. (2025). Study on the effects of scCO2 cooling and MQL conditions on the milling machinability of titanium alloys. *Wear,* Article 206036. https://doi.org/10.1016/j.wear.2025.206036
48. Şirin, E., Yıldırım, Ç. V., Şirin, Ş., Kıvak, T., & Sarıkaya, M. (2024). Comprehensive analysis of cutting temperature, tool wear, surface integrity and tribological properties in sustainable milling of Ti6Al4V alloy: LN2, nanofluid and hybrid machining. *Journal of Manufacturing Processes*, *131*, 1360–1371. https://doi.org/10.1016/j.jmapro.2024.09.120
49. Jamil, M., He, N., Zhao, W., Khan, A. M., & Laghari, R. A. (2022). Tribology and machinability performance of hybrid Al2O3 -MWCNTs nanofluids-assisted MQL for milling Ti-6Al-4 V. *The International Journal of Advanced Manufacturing Technology, 119*(3), 2127–2144. https://doi.org/10.1007/s00170-021-08279-6
50. Khanna, N., Shah, P., & Chetan (2020). Comparative analysis of dry, flood, MQL and cryogenic CO2 techniques during the machining of 15-5-PH SS alloy. *Tribology International*, *146*, 106196. https://doi.org/10.1016/j.triboint.2020.106196

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Muhammad Ahmed Khan is a doctoral candidate at the school of Mechanical Engineering, Shandong University, China. His research centers on developing high efficiency machining processes for advanced alloy, with a focus on hybrid cooling techniques, tool performance optimization, and predictive modeling of machining outcomes.

Bing Wang received the bachelor's and Ph.D. degrees in mechanical engineering from Shandong University, Jinan, China, in 2011 and 2016, respectively. He was once a joint training Ph.D. student at the Department of Mechanical Engineering, Michigan State University, East Lansing, MI, USA. From 2017 to 2020, he was a Post-Doctoral Researcher at Shandong University, and Georgia Tech Manufacturing Institute, Atlanta, GA, USA. Since 2020, he has been a Professor at the School of Mechanical Engineering, Shandong University. He has been included in the World's Top 2% Scientists consecutively in recent three years by Elsevier and Stanford University. His research interests include high-performance machining, advanced manufacturing technology, and cutting tools technology. Dr. Wang is a member of the American Society of Mechanical Engineers (ASME), Institute of Electrical and Electronic Engineers (IEEE), the Chinese Mechanical Engineering Society (CMES), and the Chinese Society of Aeronautics and Astronautics (CSAA).

Zhanqiang Liu received the B.S. and M.S. degrees in mechanical engineering from Shandong University, Jinan, China, in 1991 and 1994, respectively, and the Ph.D. degree in philosophy from the City University of Hong Kong, Hong Kong, China, in 1999. From 1999 to 2001, he was a Post-Doctoral Fellow at the School of Mechanical Engineering, Shandong University, where he has been a professor since 2002. His research interests include metal-cutting principles, cutting tools technology, monitoring of machining process, and intelligent manufacturing. Dr. Liu is a member of the American Society of Mechanical Engineers and the Executive Director of the Chinese Metal Cutting Tool Association.

Aqib Mashood Khan is a Professor at the School of Mechanical Engineering, Nanjing University of Aeronautics and Astronautics. His research focuses on metal cutting principles and cutting tool technology, with particular interest in post-processing techniques for additive manufacturing (AM). His work integrates AI, machine learning, and automation to revolutionize the post-processing phase in AM, particularly through abrasive flow machining (AFM), with applications in aerospace, automotive, and healthcare industries.

B. Mouli Prasanth is a post-doctoral candidate at the school of Mechanical Engineering, Shandong University, China. His research focuses on Electrochemical micromachining, and Non-traditional machining. His work also focuses on the enhancement of surface integrity for additive manufacturing components.

Aisheng Jiang received the bachelor's degree in mechanical engineering from Xi'an University of Technology, Xi'an, China, in 2006. Afterwards, he received the M.S. degree from Central South University, Changsha, China, in 2018, and the Ph.D. degree from Shandong University, Jinan, China, in 2025. From 2006 to 2017, he worked as an engineer at Zhuzhou Cemented Carbide Cutting Tools Co., Ltd., Zhuzhou, China. He has been a senior engineer from 2017 till now. His research interests include design, manufacture, and applications of advanced cutting tools. Dr. Jiang has published over ten peer-reviewed papers and been granted with over 100 invention patents.